

AOX target-engagement assay — plate maps

Four plates: **A15**, **B15** at 15 °C and **A27**, **B27** at 27 °C. The two temperatures use identical layouts. Every well: **5.03 mL cell suspension (OD600 0.005) → 50 µL azide slot → 20 µL inhibitor slot**. Small grey text in each well gives those two additions.

PLATE A — azide titration and the main arms at 5 mM azide

Blue ladder = azide alone, deepening with concentration. Red = SHAM, green = n-propyl gallate. Replicates are scattered so no condition sits in one row or column.

	1	2	3	4	5	6
A	TOTAL water · EtOH	AZ5+SH.2 500 mM · SHAM 50	AZ5+PG.1 500 mM · nPG 25	AZ5 500 mM · EtOH	AZ0.5 50 mM · EtOH	AZ5+SH.2 500 mM · SHAM 50
B	AZ5+PG.1 500 mM · nPG 25	TOTAL water · EtOH	AZ2 200 mM · EtOH	AZ5+SH.2 500 mM · SHAM 50	AZ10 1 M neat · EtOH	AZ5+PG.1 500 mM · nPG 25
C	AZ5 500 mM · EtOH	AZ1 100 mM · EtOH	AZ5+PG.1 500 mM · nPG 25	BLANK water · EtOH	AZ5+SH.2 500 mM · SHAM 50	TOTAL water · EtOH
D	AZ0.5 50 mM · EtOH	AZ10 1 M neat · EtOH	AZ1 100 mM · EtOH	AZ5 500 mM · EtOH	AZ2 200 mM · EtOH	BLANK water · EtOH

PLATE B — the same arms at 2 mM azide, plus high doses and inhibitor-only controls

Pale red / green = inhibitor with no azide. AZ5 repeated here so plate B can be read on its own.

	1	2	3	4	5	6
A	AZ2+SH.2 200 mM · SHAM 50	AZ5+PG1 500 mM · nPG 250	SH.2 water · SHAM 50	AZ2+PG.1 200 mM · nPG 25	AZ5+SH1 500 mM · SHAM 250	PG1 water · nPG 250
B	AZ5+SH1 500 mM · SHAM 250	AZ2+SH.2 200 mM · SHAM 50	PG.1 water · nPG 25	AZ5 500 mM · EtOH	AZ2+PG.1 200 mM · nPG 25	BLANK water · EtOH
C	AZ2+PG.1 200 mM · nPG 25	SH1 water · SHAM 250	AZ5+PG1 500 mM · nPG 250	AZ2+SH.2 200 mM · SHAM 50	PG1 water · nPG 250	AZ5+SH1 500 mM · SHAM 250
D	AZ5 500 mM · EtOH	SH.2 water · SHAM 50	AZ5+PG1 500 mM · nPG 250	PG.1 water · nPG 25	BLANK water · EtOH	SH1 water · SHAM 250

Legend

Code	Azide slot (50 μ L)	Inhibitor slot (20 μ L)	Final in vial	Code	Azide slot (50 μ L)	Inhibitor slot (20 μ L)	Final in vial
TOTAL	water	EtOH	no additions	AZ5+PG.1	500 mM	nPG 25	azide 5 + nPG 0.1
AZ0.5	50 mM	EtOH	azide 0.5 mM	AZ2+PG.1	200 mM	nPG 25	azide 2 + nPG 0.1
AZ1	100 mM	EtOH	azide 1 mM	AZ5+PG1	500 mM	nPG 250	azide 5 + nPG 1.0
AZ2	200 mM	EtOH	azide 2 mM	SH.2	water	SHAM 50	SHAM 0.2 mM
AZ5	500 mM	EtOH	azide 5 mM	SH1	water	SHAM 250	SHAM 1.0 mM
AZ10	1 M neat	EtOH	azide 10 mM	PG.1	water	nPG 25	nPG 0.1 mM
AZ5+SH.2	500 mM	SHAM 50	azide 5 + SHAM 0.2	PG1	water	nPG 250	nPG 1.0 mM
AZ2+SH.2	200 mM	SHAM 50	azide 2 + SHAM 0.2	BLANK	water	EtOH	NO CELLS — plain YMS
AZ5+SH1	500 mM	SHAM 250	azide 5 + SHAM 1.0				

Stocks

Tube	Make	Gives in vial	Tube	Make	Gives in vial
azide 50 mM	100 μ L of 1 M + 1.9 mL water	0.5 mM	SHAM 250 mM	your master, neat	1.0 mM
azide 100 mM	200 μ L of 1 M + 1.8 mL water	1 mM	SHAM 50 mM	200 μ L master + 800 μ L EtOH	0.2 mM
azide 200 mM	400 μ L of 1 M + 1.6 mL water	2 mM	nPG 250 mM	53.1 mg in 1.00 mL EtOH	1.0 mM
azide 500 mM	1.0 mL of 1 M + 1.0 mL water	5 mM	nPG 25 mM	100 μ L of 250 mM + 900 μ L EtOH	0.1 mM
azide 1 M	your stock, neat	10 mM	plain EtOH	for every no-inhibitor well	—

Cells: 88 vials $\times$ 5.03 mL $\rightarrow$ make **500 mL** at OD600 0.005 = 22.7 mL of your OD 0.11 culture + 477 mL YMS. Fill all four plates alternately from the one batch, swirling between draws. The 8 BLANK vials get plain YMS with no cells. Run 6–12 h or overnight; read **K** from the model fit, same fixed window for every vial. **Azide:** never acidify, never down the sink — collect all vials and waste in a labelled container.